

Title

Joint Handover and Trajectory Optimization in 5G-Connected UAVs

Abstract

The integration of Unmanned Aerial Vehicles (UAVs) into urban 5G and 6G cellular networks introduces complex challenges in connectivity and mobility management. Aerial user equipment installed on drones experiences unique radio propagation environments characterized by high-altitude Line-of-Sight links, sudden building blockages, and antenna sidelobe variations. To maintain performance, it is necessary to jointly optimize the UAV's physical flight trajectory and its telecommunication decisions (handovers, resource allocation).

Previous work successfully established a highly realistic, 3D ray-traced urban simulation environment and formulated a foundational Markov Decision Process (MDP) for this joint problem. However, the obtained results indicate significant room for improvement in handling the vast, multi-objective continuous action space. This research topic aims to use the established simulation framework to explore, design, and implement an advanced algorithmic solution that overcomes the limitations of the previous thesis. The primary objective is to achieve superior performance metrics: maximizing trajectory success rate and signal-to-interference-plus-noise ratio (SINR), while minimizing transmission delay, energy consumption, and redundant handovers.

Research Tasks

Task 1: Simulation frameworks study and baseline analysis

- **Task 1.1:** Inspect 3D deterministic ray-tracing simulation results in the urban grid.
- **Task 1.2:** Replicate the legacy baseline framework to extract initial performance metrics under the original MDP formulation.
- **Task 1.3:** Perform a diagnostic analysis of the previous results to identify the problems of the baseline approach (previous thesis).

Task 2: Solution exploration and algorithmic design

- **Task 2.1:** Conduct a literature review focusing on modern optimization algorithms and artificial intelligence paradigms capable of managing multi-objective, continuous action spaces in dynamic environments.
- **Task 2.2:** Formulate and propose a solution to address the joint trajectory and handover problem.
- **Task 2.3:** Map the selected methodology onto the state, action, and reward spaces; define the interfaces with the 3D environment (channel database).

Task 3: Implementation and system training

- **Task 3.1:** Implement the proposed algorithmic framework within the simulation environment.
- **Task 3.2:** Execute the training and hyperparameter tuning phases, focusing on balancing conflicting objectives such as minimizing flight time versus maintaining optimal cellular connectivity.
- **Task 3.3:** Document convergence behavior, sample efficiency, and stability metrics during the optimization process.

Task 4: Comparative evaluation and validation

- **Task 4.1:** Conduct a direct, controlled benchmark comparing the new framework against the legacy baselines (thesis, greedy, A*, etc.) across identical flight paths and destination targets.

- **Task 4.2:** Quantitatively evaluate and analyze improvements in core telecom and navigation metrics (e.g., overall latency, uplink interference, battery efficiency, and handover rate).
- **Task 4.3:** Visualize the resulting trajectories over the 3D urban grid to illustrate behavioral improvements, compiling the methodology, results, and insights into the final report.

References

Bermúdez Granados, Marina "Deep reinforcement learning-based joint handover and trajectory optimization for 5G-connected UAV", MSc Thesis, 2026

<https://upcommons.upc.edu/entities/publication/a8ce08c2-c238-4145-a5ab-5d39b12c6553>